

exec__with mask__1

April 15, 2025

An Exception was encountered at 'In [1]':

Execution using papermill encountered an exception here and stopped:

```
[1]: import numpy as np
import torch
import tensorly as tl
from tensorly.decomposition import parafac
import sparse
import pickle
import os

assert os.path.exists('sptensor.pkl'), 'No such file.'
with open('sptensor.pkl', 'rb') as f:
    data = pickle.load(f)
data = data[:, :, :, :12, :]
expected_shape = data.shape
n_components = 10

# -----
# 3. NumPy-based BPTF (Aaron's original implementation)
# -----
# Import Aaron's BPTF (which uses NumPy/sparse.COO);
# ensure that the bptf package is in your PYTHONPATH.
from bptf import BPTF as BPTF
import bptf

# Create the same data as a NumPy array and a corresponding binary mask
data_np = data
mask_np = np.zeros(expected_shape, dtype=int)
mask_np[:, :, :, 3, :] = 1
mask_np = sparse.COO(mask_np.copy())

# Instantiate and fit the NumPy-based BPTF model.
# Note: This version uses its own preprocess() function and can work with
↳ sparse.COO.
model_np = BPTF(data_shape=data_np.shape, n_components=n_components)
model_np.fit(data, mask=mask_np, max_iter=50, verbose=True, missing_val=1)
```

```

# Reconstruct using arithmetic expectation
reconstruction_np = model_np.reconstruct(mask=None, fill_value=1,
↳drop_diag=False, style='arithmetic')
frobenius_diff_np = np.sqrt(np.sum((data.todense() - reconstruction_np)**2))
print("NumPy BPTF reconstruction Frobenius norm difference:", frobenius_diff_np)

```

ITERATION 0: Time: 0.000000 Objective: -528884748.97 Change: nan

```

0%|
| 0/50 [00:00<?, ?it/s]

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0]  min uttkrp_DK = 1.634e+05
[mode 0]  min new rate = 1.634e+05   negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1]  min uttkrp_DK = 2.383e+03
[mode 1]  min new rate = 2.383e+03   negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2]  min uttkrp_DK = 3.043e+04
[mode 2]  min new rate = 3.043e+04   negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3]  min uttkrp_DK = -1.543e-08
[mode 3]  min new rate = 1.009e-01   negatives = 0
Is G_DK_M finite before updating? True

```

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 1.458e+05
[mode 4] min new rate = 1.458e+05 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

2%|
| 1/50 [00:06<05:36, 6.87s/it]

ITERATION 1: Time: 6.872252 Objective: 18832035.67 Change: 1.03561e+00

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 1.112e+05
[mode 0] min new rate = 1.112e+05 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.572e+03
[mode 1] min new rate = 1.572e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 2.296e+04
[mode 2] min new rate = 2.296e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = -1.273e-09
[mode 3] min new rate = 1.699e-06 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 1.207e+05
[mode 4] min new rate = 1.207e+05 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

4%|
| 2/50 [00:13<05:29, 6.87s/it]

ITERATION 2: Time: 6.859298 Objective: 26990853.77 Change: 4.33241e-01

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 1.029e+05
[mode 0] min new rate = 1.029e+05 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.405e+03
[mode 1] min new rate = 1.405e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.845e+04
[mode 2] min new rate = 1.845e+04 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? False
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = -2.801e-09
[mode 3] min new rate = -2.772e-09 negatives = 1
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 8.410e+04
[mode 4] min new rate = 8.410e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

6%|
| 3/50 [00:20<05:22, 6.87s/it]

ITERATION 3: Time: 6.865904 Objective: 31596269.82 Change: 1.70629e-01

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.374e+04
[mode 0] min new rate = 7.375e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.014e+03
[mode 1] min new rate = 1.014e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.220e+04
[mode 2] min new rate = 1.220e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? False
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = -1.391e-08
[mode 3] min new rate = -1.391e-08 negatives = 1
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.975e+04
[mode 4] min new rate = 4.975e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

8%|
| 4/50 [00:27<05:15, 6.86s/it]

ITERATION 4: Time: 6.854073 Objective: 32158648.08 Change: 1.77989e-02

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 4.678e+04
[mode 0] min new rate = 4.679e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 6.527e+02
[mode 1] min new rate = 6.528e+02 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 8.540e+03
[mode 2] min new rate = 8.541e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? False
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = -3.456e-10
[mode 3] min new rate = -3.454e-10 negatives = 1
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.250e+04
[mode 4] min new rate = 4.250e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

10%|

| 5/50 [00:34<05:08, 6.86s/it]

ITERATION 5: Time: 6.870105 Objective: 32565714.40 Change: 1.26581e-02

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.493e+04
[mode 0] min new rate = 7.493e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.079e+03
[mode 1] min new rate = 1.079e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.425e+04
[mode 2] min new rate = 1.425e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 5.126e-09
[mode 3] min new rate = 5.126e-09 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 5.897e+04
[mode 4] min new rate = 5.897e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

12%|
| 6/50 [00:41<05:02, 6.87s/it]

ITERATION 6: Time: 6.875293 Objective: 33256668.73 Change: 2.12172e-02

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.028e+04
[mode 0] min new rate = 7.028e+04 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.060e+03
[mode 1] min new rate = 1.060e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.446e+04
[mode 2] min new rate = 1.446e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 8.920e-09
[mode 3] min new rate = 8.922e-09 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 5.528e+04
[mode 4] min new rate = 5.528e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

14%|
| 7/50 [00:48<04:55, 6.86s/it]

ITERATION 7: Time: 6.848460 Objective: 33851754.81 Change: 1.78937e-02

Are the shape parameters positive? True
Are the rate parameters positive? True

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Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.679e+04
[mode 0] min new rate = 7.680e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.163e+03
[mode 1] min new rate = 1.164e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.595e+04
[mode 2] min new rate = 1.595e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 2.330e-08
[mode 3] min new rate = 2.330e-08 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 5.113e+04
[mode 4] min new rate = 5.113e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

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16%|

| 8/50 [00:54<04:48, 6.86s/it]

ITERATION 8: Time: 6.861861 Objective: 34270089.67 Change: 1.23578e-02

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 8.284e+04
[mode 0] min new rate = 8.284e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.199e+03
[mode 1] min new rate = 1.199e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.620e+04
[mode 2] min new rate = 1.620e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 5.787e-08
[mode 3] min new rate = 5.787e-08 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.733e+04
[mode 4] min new rate = 4.733e+04 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

18%|
| 9/50 [01:01<04:41, 6.87s/it]

ITERATION 9: Time: 6.879220 Objective: 34455157.35 Change: 5.40027e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.848e+04
[mode 0] min new rate = 7.848e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.167e+03
[mode 1] min new rate = 1.168e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.612e+04
[mode 2] min new rate = 1.612e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 6.628e-08
[mode 3] min new rate = 6.628e-08 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.483e+04
[mode 4] min new rate = 4.483e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

20%|
| 10/50 [01:08<04:34, 6.86s/it]

ITERATION 10: Time: 6.846183 Objective: 34568070.85 Change: 3.27711e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 7.857e+04
[mode 0] min new rate = 7.857e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.186e+03
[mode 1] min new rate = 1.186e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.643e+04
[mode 2] min new rate = 1.643e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 5.617e-08
[mode 3] min new rate = 5.618e-08 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.388e+04
[mode 4] min new rate = 4.388e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

22%|

| 11/50 [01:15<04:27, 6.86s/it]

ITERATION 11: Time: 6.869359 Objective: 34635900.33 Change: 1.96220e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 8.142e+04
[mode 0] min new rate = 8.143e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.209e+03
[mode 1] min new rate = 1.210e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.680e+04
[mode 2] min new rate = 1.680e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 2.364e-08
[mode 3] min new rate = 2.365e-08 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 4.333e+04
[mode 4] min new rate = 4.333e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

24%|
| 12/50 [01:22<04:20, 6.87s/it]

ITERATION 12: Time: 6.874269 Objective: 34745316.47 Change: 3.15904e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 8.426e+04
[mode 0] min new rate = 8.426e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.256e+03
[mode 1] min new rate = 1.256e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.726e+04
[mode 2] min new rate = 1.726e+04 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 1.052e-07
[mode 3] min new rate = 1.052e-07 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 3.942e+04
[mode 4] min new rate = 3.942e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

26%|
| 13/50 [01:29<04:14, 6.87s/it]

ITERATION 13: Time: 6.861155 Objective: 34859008.02 Change: 3.27214e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 9.399e+04
[mode 0] min new rate = 9.399e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.305e+03
[mode 1] min new rate = 1.305e+03 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True

Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.738e+04
[mode 2] min new rate = 1.738e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 3.038e-08
[mode 3] min new rate = 3.039e-08 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 2.624e+04
[mode 4] min new rate = 2.624e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

28%|
| 14/50 [01:36<04:07, 6.86s/it]

ITERATION 14: Time: 6.851847 Objective: 35042635.73 Change: 5.26773e-03

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 0] min uttkrp_DK = 1.136e+05
[mode 0] min new rate = 1.136e+05 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 1] min uttkrp_DK = 1.459e+03
[mode 1] min new rate = 1.459e+03 negatives = 0
Is G_DK_M finite before updating? True

Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 2] min uttkrp_DK = 1.685e+04
[mode 2] min new rate = 1.685e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 3] min uttkrp_DK = 1.577e-07
[mode 3] min new rate = 1.577e-07 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

Are the shape parameters positive? True
Are the rate parameters positive? True
Are the shape parameters finite? True
Are the rate parameters finite? True
[mode 4] min uttkrp_DK = 2.607e+04
[mode 4] min new rate = 2.607e+04 negatives = 0
Is G_DK_M finite before updating? True
Are there NaNs in G_DK_M after updating cache? False
Is G_DK_M finite after updating? True

28%|

| 14/50 [01:42<04:24, 7.35s/it]

ITERATION 15: Time: 6.858653 Objective: 34658490.29 Change: -1.09622e-02

```
-----  
AssertionError                                Traceback (most recent call last)  
Cell In[1], line 33  
    30 # Instantiate and fit the NumPy-based BPTF model.  
    31 # Note: This version uses its own preprocess() function and can work  
↪with sparse.COO.  
    32 model_np = BPTF(data_shape=data_np.shape, n_components=n_components)  
--> 33 model_np.fit(data, mask=mask_np, max_iter=50, verbose=True,  
↪missing_val=1)
```

```

    35 # Reconstruct using arithmetic expectation
    36 reconstruction_np = model_np.reconstruct(mask=None, fill_value=1,
↳drop_diag=False, style='arithmetic')

File ~/projects/bptf_new/bptf/src/bptf/bptf.py:290, in BPTF.fit(self, data,
↳mask, missing_val, **kwargs)
    286     assert is_binary(mask)
    288 self._init()
--> 290 self._update(data, mask=mask, missing_val=missing_val, **kwargs)
    291 return self

File ~/projects/bptf_new/bptf/src/bptf/bptf.py:265, in BPTF._update(self, data,
↳mask, missing_val, modes, **kwargs)
    258     print ('ITERATION %d:\t'\
    259           'Time: %f\t'\
    260           'Objective: %.2f\t'\
    261           'Change: %.5e\t'\
    262           % (itn+1, e, bound, delta))
    264 # assert delta >= 0.0
--> 265 assert delta >= -1e-2
    266 curr_elbo = bound
    267 if delta < kwargs.get('tol', 1e-4):

AssertionError:

```

```

[ ]: # -----
# 1. PyTorch-based BPTF (your version)
# -----
# Import your PyTorch-based BPTF model (adjust filename as needed)
from own_implementation import BPTF as BPTF_torch
tl.set_backend('pytorch')

device = 'cpu'
# Create a tensor from a Poisson distribution (counts) and a matching mask;
↳ensure types match
data_torch = torch.tensor(data.todense(), dtype=torch.float64, device=device)
mask_torch = torch.ones(expected_shape, dtype=torch.float64, device=device)

# Instantiate and fit the PyTorch-based BPTF model
model_torch = BPTF_torch(data_shape=expected_shape, n_components=n_components,
↳device=device)
model_torch.fit(data_torch, mask=mask_torch, max_iter=50, tol=1e-4,
↳verbose=True)
reconstruction_torch = model_torch.reconstruct(mask=mask_torch,
↳style='arithmetic')

```

```
frobenius_diff_torch = torch.norm(data_torch - reconstruction_torch, p='fro').
    ↪item()
print("PyTorch BPTF reconstruction Frobenius norm difference:",
    ↪frobenius_diff_torch)
```

```
[ ]: # -----
# 2. TensorLy CP Decomposition
# -----
# Use TensorLy's parafac for CP decomposition (same rank as n_components)
cp_decomp = parafac(data_torch, rank=n_components, n_iter_max=100,
    ↪init='random')
reconstruction_cp = tl.cp_to_tensor(cp_decomp)
frobenius_diff_cp = torch.norm(data_torch - reconstruction_cp, p='fro').item()
print("TensorLy CP decomposition Frobenius norm difference:", frobenius_diff_cp)
```